Biometric Recognition Project

(35% of the course grade)

Phase 1 — Team & Topic Selection (due October 9; required, not graded)

Form a team of 1–2 members and select one of the project topics. Submit the "Proje Takım Bildirim Formu" at https://uzak.etu.edu.tr/.

Include a one-paragraph problem statement naming:

- the problem, in your own words;
- the biometric trait(s);
- the dataset(s) you plan to use.

Phase 2 — Literature Review, System Design & Data Plan (due October 30) — 5 pts

Review at least ten sources, including recent work. From these, select a published **reference result** of the kind named in your topic, and compare your results against it.

Datasets. Use publicly available datasets. Collecting your own biometric data is not permitted without prior approval from the instructor. If any data you need requires a license agreement or a request (e.g., CASIA-Iris, IIT Delhi, IRISSEG-EP), apply immediately after topic selection: approval can take several weeks, and your data must be in hand for the checkpoint in the week of November 9. Respect the license terms of every dataset you use.

Design document. Submit a 2–3 page design document (PDF) covering:

- a short literature summary and your reference result;
- a diagram of your system and its stages;
- the dataset(s) and their licenses;
- how you will split the data, following the rule in your topic;
- how many comparisons or samples you will evaluate;
- the evaluation metrics named in your topic;
- any pretrained models, libraries or tools you plan to use.

Phase 3 — Data & Preprocessing Checkpoint (in class, week of November 9) — 4 pts

Give a 5-minute presentation covering:

- dataset statistics;
- sample preprocessing outputs;
- observed quality issues;
- samples rejected during preprocessing (e.g., failed detection or segmentation).

Upload your slides as a PDF by November 9.

Phase 4 — Development & Evaluation (no submission)

Build your system as described in your topic. Handcrafted, pretrained, fine-tuned and open-source components are all acceptable, provided they are declared.

Milestone. By November 27 you should have first results for the main metric of your topic: score distributions and the main curve named in your topic.

Evaluation rules. Your topic names the metrics, the data split and the main curve. The following rules apply to every project:

- Do not train or tune your system on the test data. If a metric needs a decision threshold, state on which data the threshold was chosen.
- People whose data were used for training or tuning must not appear in the test data. Where it is known, this includes the training data of pretrained models. A user's own enrollment samples are the only exception, and enrollment and test samples must always be different samples.
- Report error rates only at levels that your number of comparisons can support.
- Report the uncertainty of your main result, e.g., a bootstrap confidence interval computed by resampling subjects.
- Analyze your errors, and compare your results with your reference result.

Phase 5 — Final Presentation, Report & Code

Presentation with demo (in class, week of December 7) — 6 pts

Present for 10 minutes, followed by 5 minutes of questions. Upload your slides as a PDF by December 7. The session schedule will be announced.

Final report (due December 11) — 15 pts

Submit a 10–15 page PDF with the following sections:

- Introduction (problem statement)
- Literature Review
- Datasets
- Methods (the stages of your system)
- Evaluation
- Comparison with the Reference Result
- Limitations
- Conclusions
- AI Use Declaration

Code (due December 11) — 5 pts

Submit a single ZIP file containing:

- your source code;
- a README explaining how to run it;
- a requirements/environment file;
- a script that reproduces the results in your report.

Do not include in the ZIP:

- Datasets. The README must explain where and how to obtain them.
- Large model weights. Provide a download link instead.

Page Limits

References and appendices do not count toward the page limits of the design document or the final report.

Use of Generative AI Tools

You may use generative AI tools (e.g., ChatGPT, Claude, Copilot) for coding assistance, debugging, language editing and exploring ideas.

The design document and the final report must each end with a short AI Use Declaration stating:

- which tools you used;
- for which parts of the work;
- how you checked the output.

You remain fully responsible for everything you submit:

- Every cited source must be real and verified by you. AI tools frequently invent references.
- You must be able to explain any code or result in your submission. Questions during the presentation may test this.

Undeclared use of AI tools, or fabricated references, will be treated as academic misconduct.

Summary of Submissions

All submissions are made via https://uzak.etu.edu.tr/.

Deliverable	**Format**	**Due**	**Points**
Team & topic form	Online form	October 9	—
Design document	PDF	October 30	5
Checkpoint slides + presentation	PDF	November 9 (slides)	4
Final slides + presentation	PDF	December 7 (slides)	6
Final report	PDF	December 11	15
Code	ZIP	December 11	5
Total			**35**